INTERVIEW STUDY GUIDE

Sentinel

A real-time payment fraud detection platform — how it works, and why every choice was made.

This document is written to be read the night before an interview. It explains the whole system from the data up, and — more importantly — the reasoning behind each decision, phrased the way you'd defend it out loud.

0.802	**~2 ms**
PR-AUC on the future holdout	per-transaction scoring latency
0.17%	**59**
fraud rate — the core difficulty	automated tests (ML + app + e2e)

Live: sentinel-beryl-psi.vercel.app · Stack: XGBoost → ONNX · Next.js · FastAPI · Neon Postgres · Vercel

Contents

1 · The 90-second pitch

Open with what it *is*, then what makes it hard, then the outcome:

> "Sentinel is the operational system a payments company runs around a fraud model — think Stripe Radar, self-built. Every card transaction is scored the instant it arrives; the model auto-approves the ones it's confident are fine, auto-blocks the ones it's confident are fraud, and sends the uncertain middle to a human analyst with an explanation of why it was flagged."
>
> "The hard part is that fraud is only 0.17% of transactions, so accuracy is useless — a model that flags nothing is 99.83% accurate and catches zero fraud. I built the whole system around that reality: a properly evaluated gradient-boosted model at PR-AUC 0.80, served as ONNX in about two milliseconds, with a human-in-the-loop review queue, tunable decision thresholds, and per-transaction explanations — deployed full-stack on Vercel with a Postgres database."

That paragraph alone signals: you understand the domain, you know why the naive metric fails, you built a *system* (not a notebook), and you shipped it. Everything below is you being able to go three levels deeper on any sentence in it.

What makes it a "senior-looking" project, not a tutorial

- A real data pipeline with a **leakage guard** and a **quality gate** (the build fails if the model regresses).
- A clean split between **training** (heavy, offline) and **serving** (tiny, fast) — the way production ML actually works.
- Decisions framed as **business trade-offs**, not accuracy numbers.
- Deliberate **failure handling** (what happens when the model is down?) and an **audit trail**.
- It's **deployed and public**, with the demo showing genuine out-of-sample predictions.

2 · The problem, and why it's hard

Card fraud detection is a **binary classification** problem with three properties that make it unlike a textbook classifier:

1. **Extreme class imbalance.** Fraud is ~0.17% of transactions. The model sees ~580 legitimate transactions for every fraudulent one.
2. **Asymmetric, conflicting error costs.** Missing fraud costs money (a chargeback). But blocking a legitimate customer costs a sale *and* trust. You cannot minimize both at once — you trade them off.
3. **It's adversarial and time-ordered.** Fraud patterns drift; you are always predicting the future from the past.

Every design decision in the project traces back to one of these three. When an interviewer asks "why did you do X," the strongest answer usually starts with "because fraud is 0.17% of traffic…" or "because the two errors have different costs…".

> **The framing that lands:** "A fraud model doesn't make decisions — it produces a probability. Turning that probability into approve / block / review is a business decision about the relative cost of two mistakes. I made that boundary explicit and tunable, instead of hard-coding a 0.5 cutoff."

3 · The data

The **ULB credit-card fraud dataset** (OpenML #1597): 284,807 real anonymized European card transactions from 2013, of which **492 are fraud (0.173%)**.

Column	What it is	Used as a feature?
`Time`	Seconds elapsed since the first transaction in the dataset	**No** — used only to split the data (see §4)
`V1` – `V28`	PCA components — the real fields, transformed to anonymize them for privacy	Yes (28 features)
`Amount`	Transaction amount (one of only two un-transformed fields)	Yes (1 feature)
`Class`	1 = fraud, 0 = legitimate	Target label

Final feature vector: 29 dimensions (V1–V28 + Amount).

> **The honesty point (bring this up yourself — it shows integrity):** because V1–V28 are anonymized PCA components, they have no human-readable meaning. In the UI the merchant names, cities, and card numbers are *synthetic presentation metadata* — but the *scores* are computed from the real feature vectors. Interviewers know this dataset; volunteering the caveat reads as rigor, not weakness.

The demo's replay pool

The live simulator replays transactions drawn **only from the held-out test set** — data the model never trained on. This means every score shown on the live dashboard is a genuine out-of-sample prediction, not a memorized training row. Small detail, but it's the difference between an honest demo and a rigged one.

4 · The model — every decision explained

A single **XGBoost** (gradient-boosted trees) binary classifier. Final hyperparameters: depth 8, learning rate 0.05, 232 trees, `scale_pos_weight ≈ 545`, chosen by a small grid search against a chronological validation slice.

4.1 — Why gradient-boosted trees (not logistic regression, not a neural net)

vs. Logistic regression	The PCA features have non-linear interactions that a linear model can't capture. GBTs find them automatically. (LR is a fine *baseline* to mention.)
vs. Deep neural net	On ~285k rows of tabular data, gradient-boosted trees are the known state of the art — they train in seconds, need no GPU, and are far easier to explain. A neural net would be slower, hungrier for data, and harder to defend. "Right tool for tabular data" is the whole answer.
vs. Random forest	Boosting (sequential, error-correcting) generally beats bagging (parallel, variance-reducing) on this kind of problem, and XGBoost has first-class support for the imbalance knob (`scale_pos_weight`) and the PR-AUC eval metric.

4.2 — Why PR-AUC is the headline metric, not accuracy or ROC-AUC

This is **the** most important talking point in the project. Rehearse it.

- **Accuracy is a trap.** Predict "never fraud" and you're 99.83% accurate and worthless. Accuracy is dominated by the 99.83% of easy negatives.
- **ROC-AUC is flattered too.** ROC-AUC was 0.988 — impressive-looking, but it's inflated by the enormous true-negative mass. The false-positive rate barely moves because the denominator (all legit transactions) is huge.
- **Precision–Recall only looks at the rare class.** Precision = "of what we flagged, how much was really fraud." Recall = "of all fraud, how much did we catch." Neither is diluted by the easy negatives. **PR-AUC (0.802)** summarizes that curve and is the honest number.

> **Soundbite:** "ROC-AUC asks how well you rank a random fraud above a random legit transaction — easy when negatives are abundant. PR-AUC asks whether the transactions you actually flag are worth an analyst's time. For a 0.17% problem, only the second question matters."

4.3 — Why a time-based split (and why it matters more than it sounds)

I split train/test **chronologically** on `Time`: the model trains on the earlier 80% and is evaluated on the later 20% (56,962 transactions, 75 frauds).

Why not a random split? A random split lets the model train on transactions that happened *after* ones it's tested on — a subtle **data leak** that inflates every metric, because in the real world you never get to peek at the future. Fraud is deployed against the future, so it must be measured against the future. Reporting a chronological-split number is both more honest and lower (and being able to explain that is the point).

4.4 — Why `Time` is not a feature (a real bug I caught)

My first model included `Time` and scored PR-AUC 0.7995. I realized the problem: `Time` is "seconds since the dataset started," so under a chronological split **every test row's Time is larger than every training row's** — the ranges are disjoint by construction. A tree can only split on thresholds it saw in training, so at serving time every transaction falls off the same end of every Time split. The feature carries *zero* generalizable signal — only noise. Dropping it raised PR-AUC to **0.8019**.

> **This is a gift of an interview story.** It shows you reason about *why* a feature helps, not just whether a number went up. "I removed a feature and the model got better, and I can tell you exactly why" is a senior-level answer.

4.5 — Why cost-reweighting, not resampling (SMOTE/undersampling)

The imbalance is handled with XGBoost's `scale_pos_weight ≈ 545` (the ratio of negatives to positives), which tells the model to treat each fraud as ~545× more important during training.

vs. Undersampling	Throwing away 99% of the legitimate data to balance the classes discards real signal about what "normal" looks like.
vs. SMOTE (synthetic oversampling)	Inventing synthetic fraud in PCA space risks fabricating patterns that don't exist. Cost-reweighting keeps the model on the *real* data distribution and just changes how mistakes are penalized.

4.6 — The quality gate

The training pipeline is not a notebook — it has tests that **fail the build** if:

- a train/test time leak is detected (`max(train.Time) < min(test.Time)` must hold),
- holdout PR-AUC drops below 0.80,
- the exported ONNX model's predictions disagree with the trained model (parity within 1e-5; measured drift was 1.2e-7).

This is the "MLOps maturity" signal: the model can't silently regress without a red build.

5 · Turning a probability into a decision

The model outputs a probability in [0,1]. A **decision engine** maps it to an action using two thresholds:

Score range	Action	Rationale
below `t_low` (0.01)	APPROVED	Confident it's fine — no human needed.
`t_low` ≤ score < `t_high`	REVIEW	Uncertain — send to a human analyst.
≥ `t_high` (0.99)	BLOCKED	Confident it's fraud — block automatically.

Why two thresholds instead of one 0.5 cutoff

A single 0.5 cutoff forces a binary approve/block with no room for "I'm not sure." Two thresholds create a **middle band for human judgment** — the model handles the easy 99%+, humans handle the ambiguous sliver. That's the human-in-the-loop design.

Why the thresholds are tunable in the UI (the best "systems thinking" point)

Where the thresholds sit is **not a machine-learning decision — it's a business decision.** Lowering `t_high` blocks more fraud but declines more real customers. The right setting depends on the company's fraud-loss vs. customer-friction trade-off, which changes over time. So I made them a live control on the Model page, with a preview computed from the holdout table.

How the defaults were chosen (not arbitrary)

The defaults come from **operating constraints**, not a round number:

- `t_low` = the lowest threshold whose review volume stays within analyst capacity (≤0.5% of traffic).
- `t_high` = the lowest threshold where auto-blocking is ≥95% precise (so we rarely decline a real customer).

> **A subtle bug I fixed here:** my first rule was "set `t_low` so recall ≥ 90%." But the model's *ceiling* on this holdout is ~85% recall (some frauds simply score near zero). The rule was unsatisfiable and silently collapsed to `t_low = 0` — which would route *all* traffic to human review. I caught it, understood it was a real ceiling, and replaced the rule with the capacity/precision constraints above.

The operating points (memorize the shape, not the digits)

Threshold	Precision	Recall	Fraud caught (of 75)	False positives
0.01 (approve line)	0.34	0.83	62	121
0.50	0.89	0.75	56	7
0.99 (block line)	0.98	0.68	51	1

The story these numbers tell: as you raise the bar, precision climbs and recall falls — the fundamental trade-off, visible in three rows. At the block line, the model auto-blocks 51 frauds against a single false positive.

6 · Explainability — two kinds, and why both

An analyst can't act on "score: 0.94." They need *why*. The project has two explanation systems, deliberately different, each labeled for what it is.

Global — exact SHAP (on the Model page)

Computed offline during training over the holdout: the mean absolute SHAP contribution of each feature, i.e. which features drive the model *overall*. Top features: **V14, V4, V12, V11, V10** — and V14 dominating is a known signature of this dataset, which is a nice external sanity check.

Local — fast sensitivity analysis (in the review drawer)

Per transaction, at serving time: for each feature, re-score the transaction with that feature "neutralized" to its training median — "what would the score have been if this feature had been unremarkable?" The drop is that feature's impact. All 29 neutralizations are batched into **one** extra model call, so an explanation costs one inference, not 29.

Why not use SHAP for the per-transaction explanations too?

Exact SHAP is too slow for the request path. So: exact-but-slow SHAP offline for the global view, fast-but-approximate sensitivity online for the per-transaction view. I label each honestly rather than pretending the fast one is exact — which is itself a point of rigor.

7 · Serving — why ONNX, and the train/serve contract

The model is **trained** locally with the full heavy stack (XGBoost, scikit-learn, SHAP, pandas) and **exported to ONNX** — a 752 KB artifact that the serving function runs with only `onnxruntime` + `numpy`.

Why this split matters

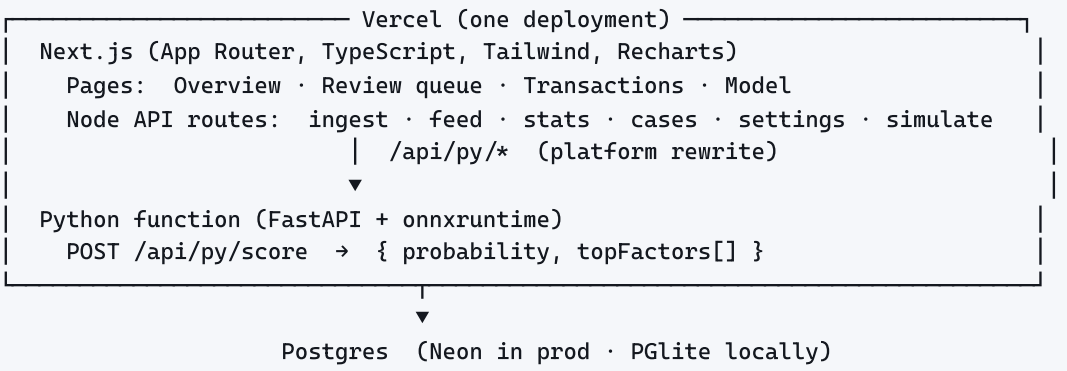
Bundle size	Shipping XGBoost + SHAP + pandas into a serverless function is hundreds of MB and slow to cold-start. ONNX + onnxruntime is tiny and boots fast.
Clean separation	Training is a heavy, occasional, offline job. Serving is a light, constant, online job. Different lifecycles, different dependencies — the industry-standard shape.
Portability	ONNX is a framework-neutral format. The serving layer doesn't care that the model came from XGBoost.

> **The safety net I'm proud of:** the export step refuses to write the ONNX file unless its predictions match the native XGBoost model within 1e-5 on 1,000 samples (measured drift: 1.2e-7). This makes "the thing I serve" provably "the thing I evaluated" — otherwise a silent conversion bug could ship a subtly different model than the metrics describe.

The fail-safe (a favorite "what if it breaks" answer)

If the scorer is unreachable, the transaction is **still recorded and routed to a human**, flagged with `scoring_error`. The policy is explicit: **never silently approve, never auto-block without a score.** An outage degrades to "everything gets human review," never to "everything sails through."

8 · System architecture & the full stack



The ingest path — the heart of the write side

Every transaction goes through one function: **score** → **decide** → **persist**, and the transaction plus (if flagged) its review case are written in a **single database transaction** so you never get an orphaned case. Two guarantees are enforced by tests:

- **Idempotency:** ingest is keyed on a client-supplied UUID, so a retry (or a double-fired simulator tick) can't duplicate a transaction, re-open a resolved case, or burn a second inference.
- **Atomicity:** transaction + case succeed together or not at all.

Notable engineering decisions

Presence-driven simulator	Vercel's free tier runs cron jobs only once a day, so a cron can't power a live feed. Instead the stream advances while someone is watching, throttled server-side (an atomic conditional `UPDATE`) so ten open tabs still produce one batch per interval — not ten. Framed as a feature: don't simulate traffic nobody is watching.
Dual database driver	One code path, two drivers chosen by whether `DATABASE_URL` is set: PGlite (in-process WASM Postgres) locally so `git clone && run` needs no database server or credentials; Neon (serverless Postgres) in production. Same schema, same migrations, same queries.
Feed cursor on a sequence, not time	Two transactions can share a timestamp; polling on time would skip or replay rows. A monotonic `seq` makes "everything after N" exact.
Case-resolution race protection	Resolving a case is a conditional update ("only if still open"), so two analysts racing on the same case get a 409 instead of silently overwriting each other.
Raw features never leave the server	The 29-dim vector is stripped from every API response — it's meaningless to a human and pure payload weight.

Testing (the "I verify my work" signal)

59 automated tests across three layers: 16 Python (leakage guard, PR-AUC floor, ONNX parity, scorer behavior), 39 TypeScript (decision engine boundaries, fail-safe ingest, idempotency, API routes with a real in-memory Postgres), and 4 Playwright end-to-end tests that drive the real app + scorer + database in a browser.

9 · Deployment, and the bugs it exposed

Deployed to Vercel as a single project: the Next.js app and the Python scorer ship together, with Neon Postgres provisioned through Vercel's marketplace. Three problems surfaced only in production — and being able to tell these stories is worth as much as the clean parts.

Bug 1 — The build booted a database it didn't need.

`next build` imports every route to collect metadata, and my database module created its connection eagerly at import — so the build tried to boot PGlite in a worker with no database, and crashed. **Fix:** made the connection lazy (a Proxy that connects on first query), so importing the module is free.

Bug 2 — Every score came back empty in production.

The internal Node→Python call used `VERCEL_URL`, which points to the *deployment* URL. That URL has SSO "Deployment Protection" on, so the self-call was redirected to a login page and failed — every transaction hit the fail-safe and went to review with no score. **Fix:** pointed the internal call at the public production domain via an env var. (Good story: the fail-safe *worked* — an outage degraded to human review, exactly as designed.)

Bug 3 — First deploy rejected: file over 100 MB.

The 300 MB raw dataset was being uploaded. It's needed only to train locally; the committed model artifacts are what deploy. **Fix:** a `.vercelignore` excluding the raw data, local database, and training scripts.

> **Also worth mentioning:** PGlite is single-writer. During testing I ran several processes against the same local database at once and corrupted it ("RuntimeError: Aborted"). Lesson: know your tool's concurrency model. Production uses Neon, which is unaffected.

10 · "Why X over Y" — master decision table

The single most useful page for rapid review. Each row is a question you might be asked.

Decision	Chosen	Over	Because
Model family	XGBoost	Logistic reg / neural net	State of the art on tabular data; captures non-linear interactions; fast, no GPU, explainable.
Headline metric	PR-AUC	Accuracy / ROC-AUC	Only PR focuses on the rare class; the others are flattered by the 99.83% easy negatives.
Train/test split	Time-based	Random	Random leaks the future into training and inflates metrics; fraud is deployed against the future.
`Time` feature	Excluded	Included	Disjoint train/test ranges under a time split → no generalizable signal. Removing it raised PR-AUC.
Imbalance handling	Cost reweighting	SMOTE / undersampling	Keeps the real data distribution; no discarded signal, no fabricated fraud.
Decision boundary	Two tunable thresholds	One fixed 0.5 cutoff	Creates a human-review band; the cutoff is a business trade-off, not a constant.
Serving format	ONNX	Ship XGBoost	Tiny bundle, fast cold start, clean train/serve separation; parity-checked against the trained model.
Local explanations	Sensitivity analysis	Exact SHAP	SHAP is too slow for the request path; sensitivity is one batched call and honestly labeled as approximate.
Scorer-down behavior	Route to human	Approve / block by default	Fail safe: never silently approve, never auto-block without evidence.
Local database	PGlite	Docker Postgres	Zero-setup `clone & run`; same SQL as prod via one dual-driver module.
Prod database	Neon serverless	Traditional Postgres	Serverless-friendly connection model; free tier; native Vercel integration.
Live traffic	Presence-driven	Cron job	Free-tier cron is daily-only; also, don't burn compute simulating traffic nobody's watching.
Hosting	All-on-Vercel	Split frontend/backend	One repo, one deploy, one URL; Vercel runs Python too, so the model lives beside the app.

11 · Anticipated interview questions

Q. Your PR-AUC is 0.80 — is that good?

A. For this dataset with an honest *time-based* split, yes — random-split numbers you see online (often 0.85+) are optimistic because they leak future data. I deliberately report the harder, more realistic number. And "good" depends on the operating point: at the block threshold I get 98% precision, which is what actually matters for auto-blocking.

Q. How would you improve the model?

A. The biggest lever isn't the algorithm — it's features. This dataset gives me anonymized PCA components; in a real system I'd engineer behavioral features: velocity (transactions per card per hour), amount relative to the card's history, geographic and merchant-category anomalies, time-since-last-transaction. Those encode the patterns fraud actually follows. After that: probability calibration, and a threshold that adapts per segment.

Q. How do you know the model isn't overfitting?

A. It's evaluated on 56,962 transactions it never saw, drawn from a later time period than training — the hardest honest test. I also early-stop on a validation slice and keep the tree depth bounded. The gap between validation and holdout PR-AUC is small.

Q. What happens when fraud patterns change (drift)?

A. Two things. Short term, the tunable thresholds let an operator react without retraining. Long term, the analyst approve/block decisions are stored — they're new labels. That's a retraining flywheel: the human-in-the-loop system generates its own training data. Drift monitoring (comparing live score distributions to training) is my top "next feature."

Q. Why serverless / why Vercel for an ML app?

A. Because I separated training from serving. Serving a fixed tree ensemble is a light, stateless, bursty workload — a perfect fit for serverless. Training is offline and doesn't touch production. If the model were a large deep net needing a GPU, I'd serve it differently (a dedicated inference service), and I can talk about that trade-off.

Q. How would this scale to millions of transactions a day?

A. The scorer is stateless and horizontally scalable — that's the easy part. The bottlenecks would be the database write path and the analyst queue. I'd move ingest to an async queue, batch DB writes, add read replicas for the dashboards, and cache aggregates. The 2 ms scoring latency itself has plenty of headroom.

Q. Walk me through what happens when a card is swiped.

A. The transaction hits the ingest function → it calls the Python scorer over HTTP → the ONNX model returns a probability plus the top contributing factors → the decision engine compares the probability to the two thresholds → the transaction (and a review case if it's in the middle band) is written to Postgres in one atomic transaction → the dashboard picks it up on its next poll. End to end, a few milliseconds of actual compute.

Q. What was the hardest part?

A. Honestly, the modeling judgment calls, not the code. Realizing that `Time` couldn't generalize under a time split, and that my "90% recall" threshold rule was mathematically unsatisfiable given the model's ceiling — both required understanding *why* the numbers were what they were, not just reading them off.

12 · Honest weaknesses & what's next

Volunteering limitations signals maturity. Have two or three ready.

- **Anonymized features cap realism.** The PCA components mean I can't do the behavioral feature engineering that drives real fraud systems. I'd want raw transaction fields.
- **No probability calibration.** XGBoost scores aren't guaranteed to be true probabilities. For "expected loss" math you'd want Platt scaling or isotonic calibration.
- **Single model, no champion/challenger.** Production fraud systems A/B test models and thresholds; I have the registry table for it but not the experiment framework.
- **No automated retraining.** The analyst-label flywheel exists in the schema but isn't wired to a retraining job yet.
- **Drift is unmonitored.** My top next feature: compare live score distributions against training to detect when the world has moved.

> **Roadmap, one line:** behavioral features → probability calibration → drift monitoring → retrain-from-analyst-labels → champion/challenger threshold experiments.

13 · One-page cheat sheet

Thing	Number / answer
What it is	Real-time fraud detection platform: score → auto-decide → human review
Dataset	ULB credit-card, 284,807 txns, 492 fraud (0.173%)
Model	XGBoost, depth 8, 232 trees, cost-reweighted (~545×)
Features	29 (V1–V28 + Amount); Time excluded on purpose
Headline metric	PR-AUC 0.802 (ROC-AUC 0.988) on a time-based holdout
Why PR-AUC	Only it focuses on the 0.17% rare class; accuracy/ROC are flattered
Decision	Two tunable thresholds → approve / review / block
Default thresholds	0.01 (approve line) / 0.99 (block line) — from capacity & precision constraints
At block line	98% precision, 51/75 fraud caught, 1 false positive
Serving	ONNX (752 KB), ~2 ms, parity-checked to 1.2e-7
Explainability	Global SHAP (offline) + per-txn sensitivity (online); top feature V14
Fail-safe	Scorer down → route to human; never silent-approve or blind-block
Stack	Next.js + FastAPI + Neon Postgres, all on Vercel; PGlite locally
Tests	59 total: 16 pytest + 39 vitest + 4 Playwright e2e
Top 3 stories	(1) dropped Time, model improved; (2) fail-safe caught a prod outage; (3) threshold rule was unsatisfiable

Written as an interview-prep companion to the Sentinel project. Live demo: sentinel-beryl-psi.vercel.app · All figures pulled from the trained model's committed metrics.